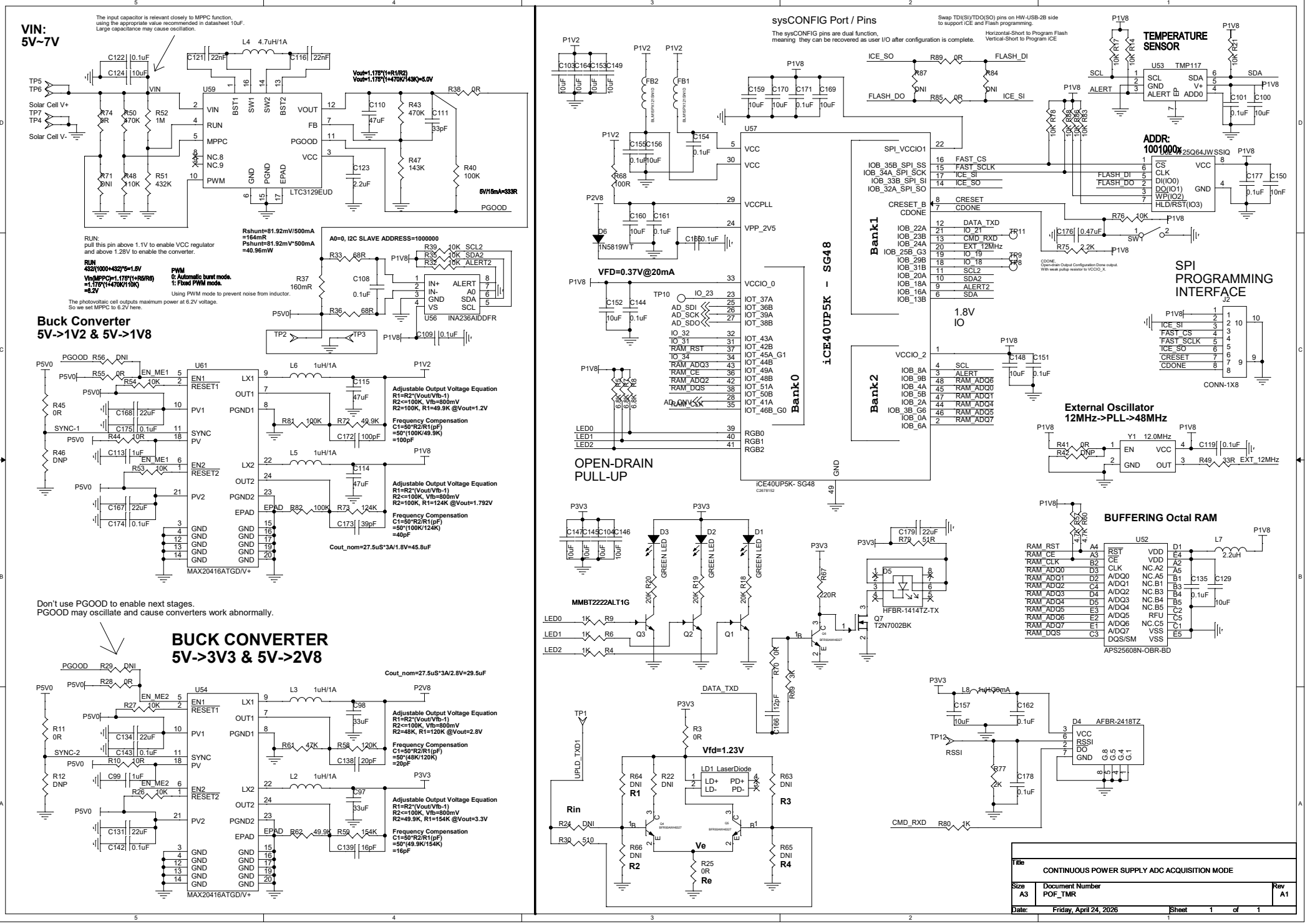
VIN:
5V~7V
The input capacitor is relevant closely to MPPC function, using the appropriate value recommended in datasheet 10uF. Large capacitance may cause oscillation.
Solar Cell V+
Solar Cell V-
LTC3129EUD
Vout=1.175*(1+R1/R2)
Vout=1.175*(1+470K/143K)=5.0V
5V/15mA=333R
PGOOD
RUN:
pull this pin above 1.1V to enable VCC regulator and above 1.28V to enable the converter.
RUN
432/(1000+432)*5=1.5V
Vin(MPPC)=1.175*(1+R5/R6)
=1.175*(1+470K/110K)
=6.2V
PWM
0: Automatic burst mode.
1: Fixed PWM mode.
Using PWM mode to prevent noise from inductor.
The photovoltaic cell outputs maximum power at 6.2V voltage. So we set MPPC to 6.2V here.
Rshunt=81.92mV/500mA
=164mR
Pshunt=81.92mV*500mA
=40.96mW
A0=0, I2C SLAVE ADDRESS=1000000
INA236AIDDFR
Buck Converter
5V->1V2 & 5V->1V8
MAX20416ATGD/V+
Adjustable Output Voltage Equation
R1=R2*(Vout/Vfb-1)
R2<=100K, Vfb=800mV
R2=100K, R1=49.9K @Vout=1.2V
Frequency Compensation
C1=50*R2/R1(pF)
=50*(100K/49.9K)
=100pF
Adjustable Output Voltage Equation
R1=R2*(Vout/Vfb-1)
R2<=100K, Vfb=800mV
R2=100K, R1=124K @Vout=1.792V
Frequency Compensation
C1=50*R2/R1(pF)
=50*(100K/124K)
=40pF
Cout_nom=27.5uS*3A/1.8V=45.8uF
Don't use PGOOD to enable next stages.
PGOOD may oscillate and cause converters work abnormally.
BUCK CONVERTER
5V->3V3 & 5V->2V8
Cout_nom=27.5uS*3A/2.8V=29.5uF
Adjustable Output Voltage Equation
R1=R2*(Vout/Vfb-1)
R2<=100K, Vfb=800mV
R2=48K, R1=120K @Vout=2.8V
Frequency Compensation
C1=50*R2/R1(pF)
=50*(48K/120K)
=20pF
Adjustable Output Voltage Equation
R1=R2*(Vout/Vfb-1)
R2<=100K, Vfb=800mV
R2=49.9K, R1=154K @Vout=3.3V
Frequency Compensation
C1=50*R2/R1(pF)
=50*(49.9K/154K)
=16pF
sysCONFIG Port / Pins
The sysCONFIG pins are dual function, meaning they can be recovered as user I/O after configuration is complete.
Swap TDI(SI)/TDO(SO) pins on HW-USB-2B side to support ICE and Flash programming.
Horizontal-Short to Program Flash
Vertical-Short to Program ICE
TEMPERATURE SENSOR
TMP117
ADDR:
1001000x
25Q64JWSSIQ
VFD=0.37V@20mA
iCE40UP5K - SG48
Bank0
Bank1
Bank2
1.8V
IO
SPI
PROGRAMMING
INTERFACE
CONN-1X8
External Oscillator
12MHz->PLL->48MHz
OPEN-DRAIN
PULL-UP
MMBT2222ALT1G
HFBR-1414TZ-TX
T2N7002BK
BUFFERING Octal RAM
APS25608N-OBR-BD
Vfd=1.23V
LD1 LaserDiode
AFBR-2418TZ
Rin
Ve
Re
R1
R2
R3
R4
CONTINUOUS POWER SUPPLY ADC ACQUISITION MODE
Size
A3
Document Number
POF_TMR
Rev
A1
Date:
Friday, April 24, 2026
Sheet
1
of
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